

OBJECT ORIENTED PROGRAMMING

1. Control Structures

- ☐ Sequential Control
- ☐ Decision Control (if, if-else, switch-case)
- ☐ Loop Control (for, while, do-while)
- ☐ Branching Statements (break, continue, return, goto)

2. Functions

- ☐ Defining and Declaring Functions
- ☐ Function Overloading
- ☐ Inline Functions
- ☐ Recursive Functions
- ☐ Passing Arguments (Pass by Value, Pass by Reference, Pass by Pointer)
- ☐ Default Arguments

3. Arrays

- ☐ Single-Dimensional Arrays
- ☐ Multi-Dimensional Arrays
- ☐ Passing Arrays to Functions
- ☐ Arrays of Pointers

4. Pointers

- ☐ Basics of Pointers
- ☐ Pointer Arithmetic
- ☐ Pointers to Arrays
- ☐ Pointers to Functions
- ☐ Multi-Dimensional Pointers as Arrays

5. Objects and Classes

- ☐ Defining a Class
- ☐ Defining Objects
- ☐ Member Variables and Member Functions
- ☐ Defining Functions Inside the Class
- ☐ Defining Functions Outside the Class
- ☐ Access Specifiers: private, public, protected

6. Constructors and Destructors

- ☐ Constructor
 - ☐ Default Constructor (Constructor with No Arguments)
 - ☐ Constructor with One Argument (Type-cast Constructor)
 - ☐ Constructor with Default Arguments
- ☐ Destructor

7. The **this** Pointer

- ☐ Use of **this** Pointer
- ☐ Resolving Naming Conflicts with **this**

8. Differences

- ☐ Difference between Class and Struct

9. Data Abstraction and Encapsulation

- ☐ Data Abstraction through Encapsulation
- ☐ Setter/Mutator and Getter/Accessor Methods
- ☐ Access Methods

10. Copying Objects

- ☐ Shallow Copy
- ☐ Deep Copy

11. Const Methods

- ☐ Definition and Use of Const Methods

12. Composition

- ☐ Concept and Examples of Composition

13. Friend Functions and Classes

- ☐ Friend Functions
 - ☐ Efficient but can compromise Encapsulation
- ☐ Friend Classes

14. Operator Overloading

- ☐ Overloading Arithmetic Operators
- ☐ Overloading Arrow Operator ($\rightarrow$)
- ☐ Overloading Parenthesis Operator ($()$)
- ☐ Overloading Comma Operator ($,$)
- ☐ Overloading Unary Operators
- ☐ Overloading Binary Operators

15. Inheritance

- ☐ Concept of Inheritance
- ☐ Types of Inheritance
 - ☐ Single Inheritance
 - ☐ Multiple Inheritance
 - ☐ Multi-level Inheritance
 - ☐ Hierarchical Inheritance
 - ☐ Hybrid Inheritance
- ☐ Direct and Indirect Base Class Relationships

16. Containment in OOP

- ☐ Composition
- ☐ Container
- ☐ Iterator
- ☐ Proxy
- ☐ Association

17. Polymorphism

- ☐ Virtual Functions
- ☐ Dynamic Binding
- ☐ Concept of vPointer and vTable

18. File Handling

- ☐ File Input and Output Streams
- ☐ File Opening and Closing
- ☐ Reading from and Writing to Files
- ☐ File Modes